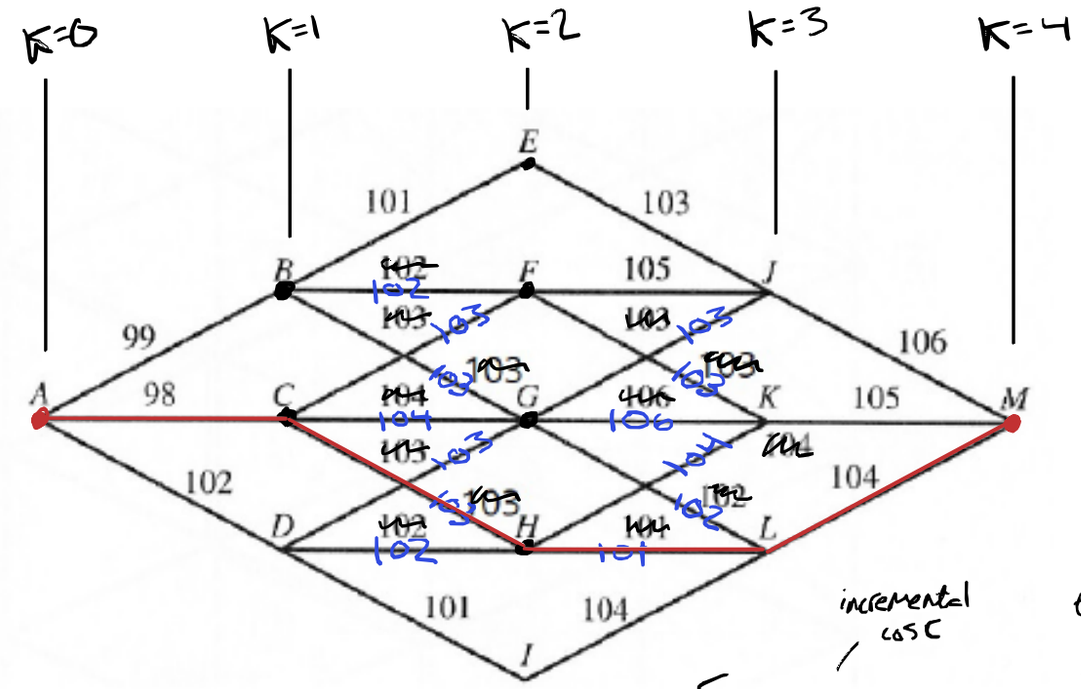
ECE675
HW3
Problem 2
Dynam. Prog.



*optimal path!

$$V_k = \min_{u \in U} \{ \underbrace{L(k,x,u)}_{\text{incremental cost}} + \underbrace{V_{k+1}(F(k,x,u))}_{\text{terminal cost}} \}$$

@ K=4

$V_4(M) = 0 + 0$ (no terminal cost)

@ K=3

$V_3(J) = 106 \quad V_3(K) = 105 \quad V_3(L) = 104$

@ K=2

$V_2(E) = 103 + V_3(J) = 209$

$V_2(F) = \min\{105 + V_3(J), 103 + V_3(K)\} = 208$

$V_2(G) = \min\{103 + V_3(J), 106 + V_3(K), 102 + V_3(L)\} = 206$

$V_2(H) = \min\{104 + V_3(K), 101 + V_3(L)\} = 205$

$V_2(I) = 104 + V_3(L) = 208$

@ K=1

$V_1(B) = \min\{101 + V_2(E), 102 + V_2(F), 103 + V_2(G)\} = 309$

$V_1(C) = \min\{103 + V_2(F), 104 + V_2(G), 103 + V_2(H)\} = 308$

$V_1(D) = \min\{103 + V_2(G), 102 + V_2(H), 101 + V_2(I)\} = 307$

@ K=0

$V_0(A) = \min\{99 + V_1(B), 98 + V_1(C), 102 + V_1(D)\} = 406$

- Minimum cost = 406
- Best route highlighted red